An LSL-Middleware Prototype for VR/AR Data Collection

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ABSTRACT

Multimodal data allows great opportunity in research and interaction in virtual/augmented reality (VR/AR) experiences for measuring human behavior. However, it is challenging to collect, coordinate, and synchronize high-volume of data while preserving the high frame-rate and the quality. Lab Streaming Layer (LSL) is an open-source framework that allows various types of multimodal data to be synchronously collected. In this work, we push the boundaries of the LSL framework by introducing an open-Source MultimOdal framewOrk for Tracking Hardware (SMOOTH)—an LSL-based middleware. The SMOOTH on top of the LSL framework supports real-time data collection and streaming using VR/AR hardware, which the LSL does not currently support, such as Microsoft Azure Kinect and Windows Mixed Reality headset and controllers. We also conducted a preliminary evaluation to understand the effectiveness of the SMOOTH and how well it performed on the task of collecting synchronized image, depth, infrared, audio, and 3D motion tracking data qualitatively.

CCS CONCEPTS

• **Information systems** → **Multimedia streaming**; • **Human-centered computing** → *Mixed / augmented reality*; *Virtual reality*.

KEYWORDS

Lab Streaming Layer, Microsoft Azure Kinect, Windows Mixed Reality, Virtual Reality, Motion Data Collection, Multimodal Data for Virtual Reality.

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1 INTRODUCTION

As a variety of sensing and controlling devices have been increasingly proposed and invented in the field of VR/AR, research and applications involve various types of multimodal data from heterogeneous devices—e.g., data related to users' emotions, behavior, and physiological signals. Comprehensive analysis of these data can lead us to understand better the users' perception/cognition process and their experience in VR/AR; thus, reliable data collection and sharing mechanisms are essential for accurate and robust evaluations for VR/AR research and applications.

The open-source *Lab Streaming Layer (LSL)* [1] has gained much attention from researchers as a system for the unified collection of measurement time series in research experiments. This handles the networking, time-synchronization, and (near-) real-time access as well as optionally the centralized collection, viewing and disk recording of the data. Since the LSL was initially developed to collect neurological brain signals, collecting physiological signals, such as electroencephalogram (EEG) data, is one of the most common uses for LSL [2]. Wunderlich and Gramann used LSL in a project that required an extension of LSL to support EEG devices, which was not supported previously [3].

We performed a systematic literature review to investigate the use of LSL and identify potential barriers. Our review analysis showed an increasing trend in the use of LSL for VR/AR-related research; however, the purpose of the use is mainly for neurological brain signals, despite the apparent benefit collecting motion data during VR/AR experience. These findings encouraged us to design and develop an easy-to-use open-source middleware called *open-Source MultimOdal framewOrk for Tracking Hardware (SMOOTH)*, which novice VR/AR researchers and students can easily adapt in their systems to collect and stream data from Microsoft Azure Kinect and the Windows Mixed Reality headsets and controllers.

2 SMOOTH DEVELOPMENT

Windows Mixed Reality (MR) headset and controllers; and Microsoft (MS) Azure Kinect, are not currently supported by LSL but are one of the most frequently used devices in the field. Our implementation was based on Unity (version 2019.2.17f1) and Visual Studio 2019 (version 16.6.4). Overall how-to-use instructions and code files are publicly available on Github[1].

[1]https://github.com/HCI-UD/LSL-Extensions.git

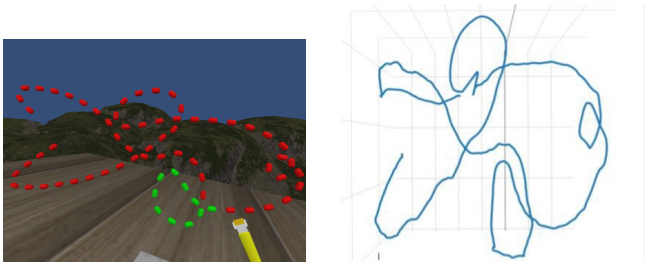


Figure 1: A drawing task with a fish model for motion data collection (left). The line tracing is visualized in 3D (right).

2.1 Preliminary Evaluation

We conducted a preliminary evaluation for our SMOOTH prototype examining the quality of our data collection results. All the evaluations were performed using one machine (OMEN by HP Obelisk Desktop 875-0024 with Intel (R) Core (TM) i7-8700 CPU @ 3.20 GHz, 16.0 GB RAM, 2TB HDD @ 7200 rpm, 256 GB SSD, and NVIDIA GeForce RTX 2080 GPU).

Windows MR. For the evaluation in the data collection of Windows MR controllers via SMOOTH middleware, we analyzed the collected data by using a VR line drawing application in Windows MR. We implemented Python programs to receive, record, and save the data from Windows MR into csv files. Multiple users with the MR headset were requested to trace the shape of a fish displayed in the view multiple times, using their dominant hand. The motion data of the controller were collected and visualized for any data loss using R Studio (see a test example in Figure 1). The visualization shows that there is no noticeable data loss with the data captured by SMOOTH.

Azure Kinect. To evaluate the Azure Kinect data collection, we implemented programs that use the Numpy[2] libraries to arrange the image data into the proper shape and format. For the color images, this involved collecting the blue, green, red, alpha channels, and rearranging them to RGBA format. Similarly for the depth images, rearranging the one-dimensional array of points into a two-dimensional shape was required. The 16-bit data were scaled into an 8-bit range so the image could be shown in grayscale. Once the image data were arranged properly, the Pillow library[3] was used to save the data as image files. The skeleton data and audio data programs saved the raw numbers into csv files, and we designed a program to synchronize the color images and skeleton data to show the skeleton tracking overlaid on top of the color images. Examples of the two independent camera images are shown in Figure 2. All these images appeared to be of the same quality and resolution as those obtained using the SDK without any noticeable delays.

The Kinect interface allowed four different parameters to be varied: color image resolution, color image format, infrared image resolution, and frame rate. In order to keep consistency between the experiments, the color image was fixed in the BGRA-32 format.

[2]https://numpy.org/
[3]https://pillow.readthedocs.io/en/stable/

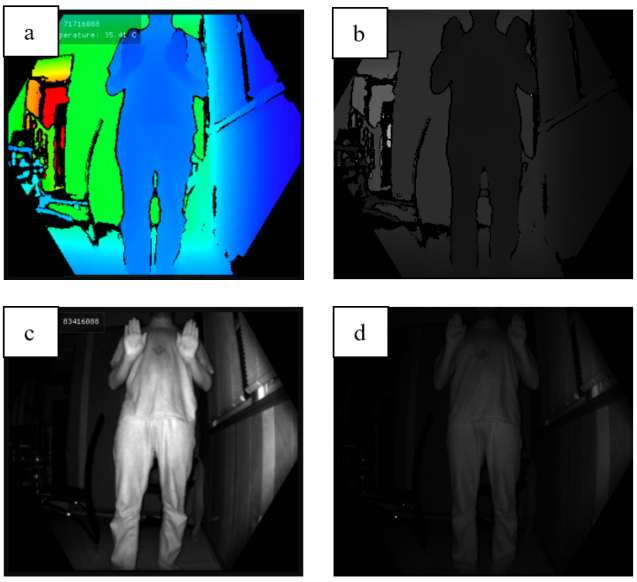


Figure 2: Examples of depth images collected from Azure Kinect SDK (a) and our SMOOTH prototype (b). Examples of infrared images collected from Azure Kinect SDK (c) and our SMOOTH prototype (d). The visual difference between the images is due to the data conversion.

3 CONCLUSIONS

In this work, we developed our in-house open-source middleware, SMOOTH, to expand multimodal data collection of LSL in VR/AR developments. The SMOOTH uses LSL as its backbone, but offers more support for data collection for tracking devices. In particular, it allows data collection from both an HMD (and its controllers) and MS Azure Kinect, which can track the user's body, on different development environments, e.g., Unity and Visual Studio. This work was a preliminary effort towards supporting several new hardware gadgets for VR/AR, and like any other works, had some limitations. In the future, we will extend our SMOOTH to support other sensing or tracking devices, including MS HoloLens and Oculus Quest, and conduct more comprehensive comparative benchmark evaluations using other existing data collection frameworks.

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REFERENCES

[1] Christian Kothe. [n.d.]. Lab Streaming Layer (LSL) - A software framework for synchronizing a large array of data collection and stimulation devices. https://github.com/sccn/labstreaminglayer. [Accessed on 2021-06-08].
[2] Hakim Si-Mohammed, Catarina Lopes-Dias, Maria Duarte, Ferran Argelaguet, Camille Jeunet, Géry Casiez, Gernot Müller-Putz, Anatole Lécuyer, and Reinhold Scherer. 2020. Detecting system errors in virtual reality using EEG through error-related potentials. In *Proceedings of the IEEE Conference on Virtual Reality and 3D User Interfaces*. 653–661.
[3] Anna Wunderlich and Klaus Gramann. 2020. Brain dynamics of assisted pedestrian navigation in the real-world. *bioRxiv* (2020).